Exploring multiverse analyses using decision trees

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Michael S. Truong
York University

Gabriel Crone
York University

Ji Yeh Choi
Canadian University Dubai

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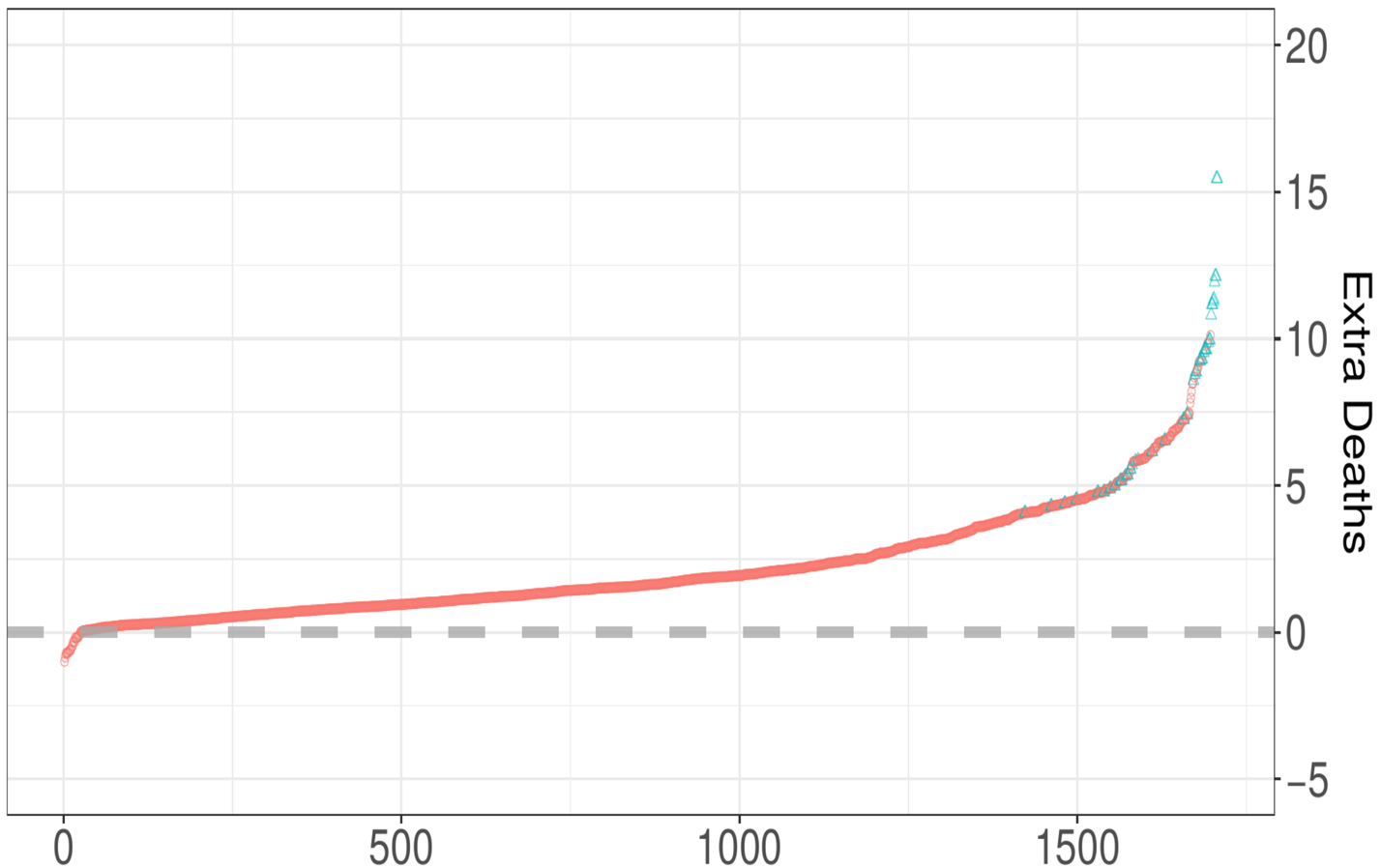
emstruonger@gmail.com

1 Main Point

1.1 The Problem of Multiverse Analyses and Their Interpretation

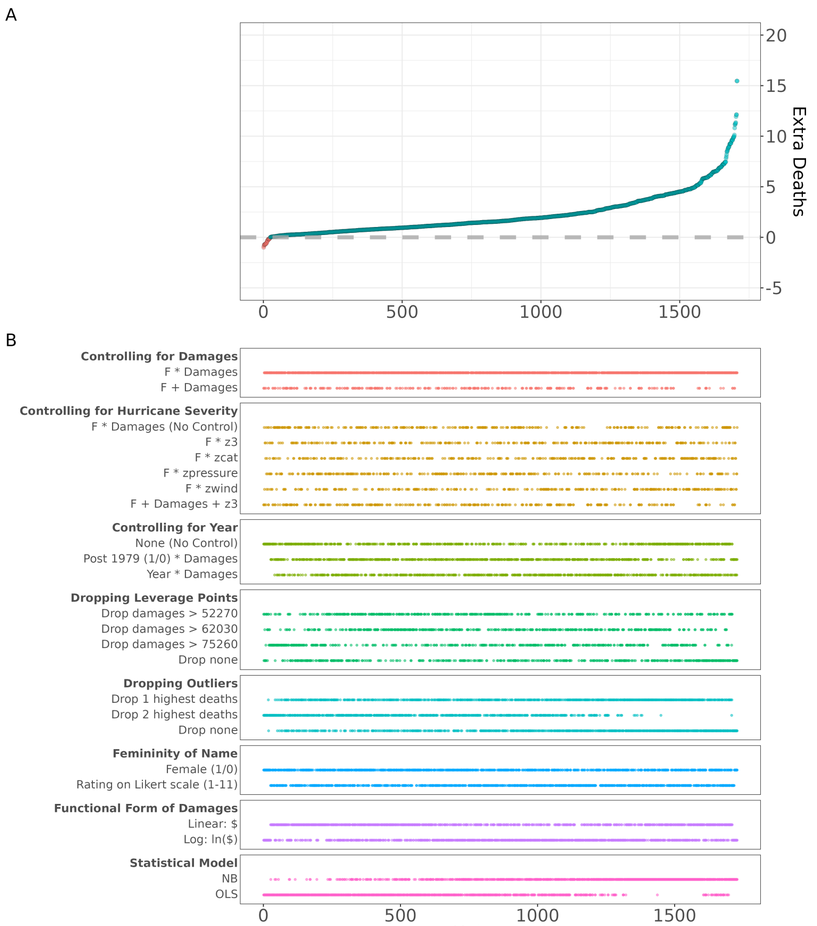
Discomfort: Can you '*follow the rules of statistics*', but also get '*any result you want*'?

- Multiverse papers
- Multi-analyst papers

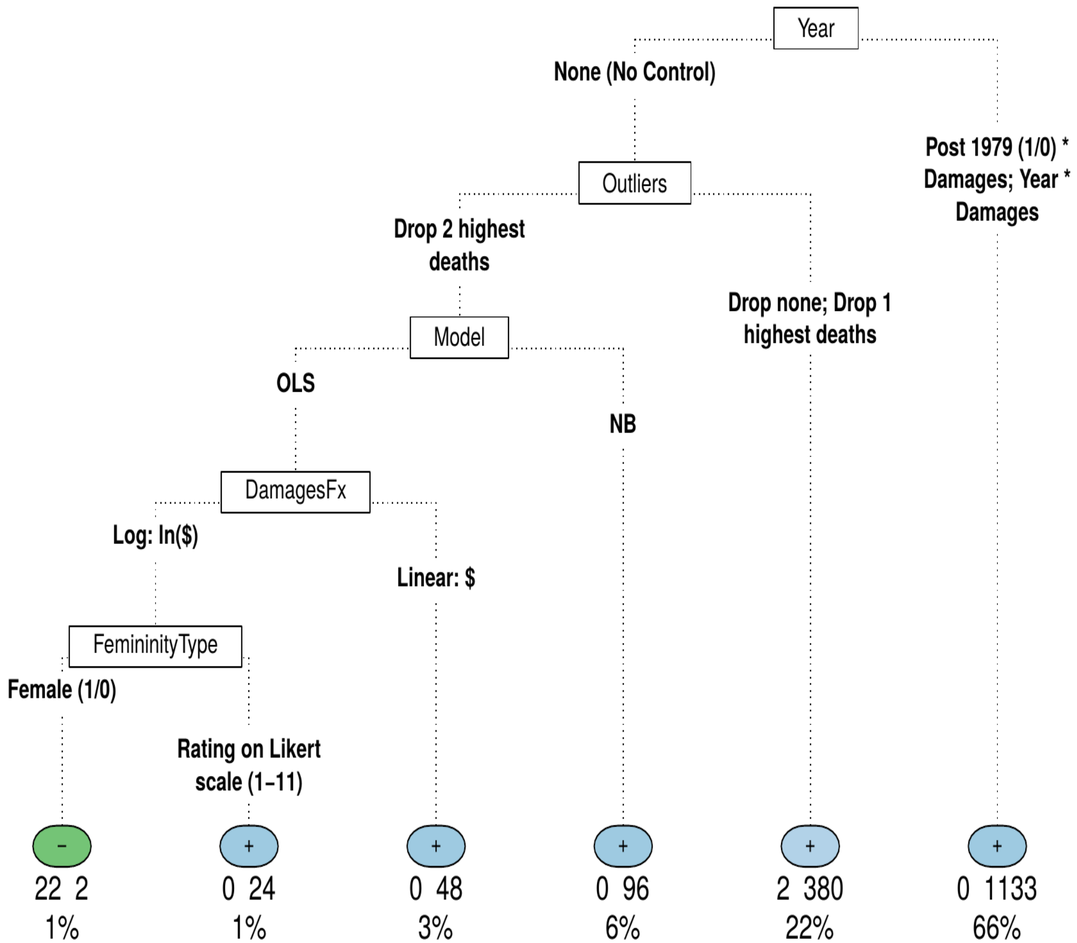


Specification Curve Simonsohn et al. (2020)

1.2 Interpretable Multiverse Analyses



Specification Curve Simonsohn et al. (2020)

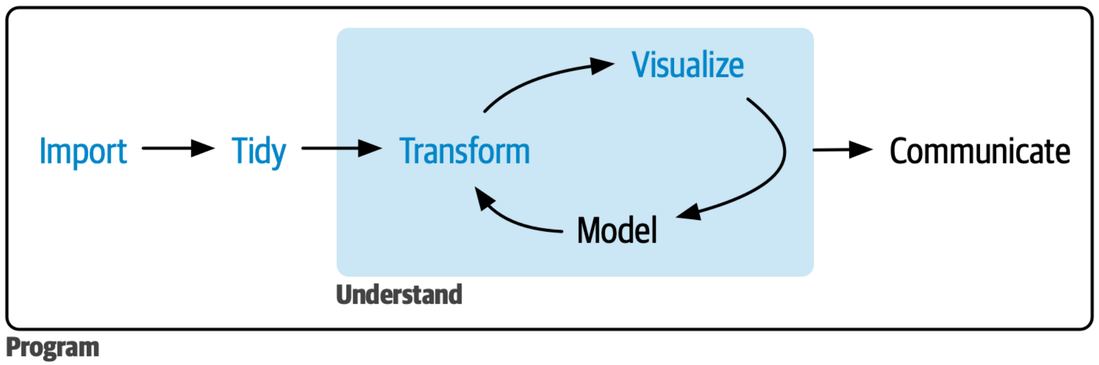


Correctly Predicts Sign of Result 99.77% of the Time

2 What is a multiverse analysis?

- What happens in a typical analysis?
- What is an example of a multiverse analysis?

2.1 What happens in a typical analysis?



General workflow from Wickham et al. (2023)

- Typical Analysis (Steegen et al. 2016):
  - Many arbitrary decisions
    - Controls?
    - Variable transformations?
    - Outliers?
    - Missing data?
    - What estimator?
  - Question: **What effect do these decisions have on your statistical results?**

Instead, Multiverse Analyses:

- **All possible and reasonable combinations of choices** (Steegen et al. 2016; Del Giudice and Gangestad 2021)

2.2 What is an example of a multiverse analysis?

Claims:

> "We use more than six decades of death rates from US hurricanes to show that feminine-named hurricanes cause significantly more deaths than do masculine-named hurricanes."

> "…our model suggests that changing a severe hurricane's name from Charley (MFI = 2.889, 14.87 deaths) to Eloise (MFI = 8.944, 41.45 deaths) could nearly triple its death toll."

- Their explanation: Stereotype that women are not as deadly as men → people don't adequately prepare for feminine vs. masculine-named hurricanes



Female hurricanes are deadlier than male hurricanes

Kiju Jung[a,1], Sharon Shavitt[a,b,1], Madhu Viswanathan[a,c], and Joseph M. Hilbe[d]

[a]Department of Business Administration and [b]Department of Psychology, Institute of Communications Research, and Survey Research Laboratory, and [c]Women and Gender in Global Perspectives, University of Illinois at Urbana–Champaign, Champaign, IL 61820; and [d]Department of Statistics, T. Denny Sanford School of Social and Family Dynamics, Arizona State University, Tempe, AZ 85287-3701

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Do people judge hurricane risks in the context of gender-based expectations? We use more than six decades of death rates from US hurricanes to show that feminine-named hurricanes cause significantly more deaths than do masculine-named hurricanes. Laboratory experiments indicate that this is because hurricane names lead to gender-based expectations about severity and this, in turn, guides respondents' preparedness to take protective action. This finding indicates an unfortunate and unintended consequence of the gendered naming of hurricanes, with important implications for policymakers, media practitioners, and the general public concerning hurricane communication and preparedness.

gender stereotypes | implicit bias | risk perception | natural hazard communication | bounded rationality

Estimates suggest that hurricanes kill more than 200 people in the United States annually, and severe hurricanes can cause fatalities in the thousands (1). As the global climate changes, the frequency and severity of such storms is expected to increase (2). However, motivating hurricane preparedness remains a major challenge for local and state authorities (3). Although natural hazards such as hurricanes represent both physical and social phenomena (4, 5), meteorologists and geoscientists point out that too little attention has been paid to findings from the social sciences about subjective risk perceptions (6, 7). Those findings highlight the importance of understanding how assessments of risk from threats in the environment are often influenced not only by environmental and social cues (8, 9), but also by irrelevant psychological factors (10–12).

We demonstrate that a natural disaster can, merely by being symbolically associated with a given sex through its assigned name, be judged in ways congruent with the corresponding social roles and expectations of that sex (13–16). In particular, analyses of archival data on actual fatalities caused by hurricanes in the United States (1950–2012) indicate that severe hurricanes with feminine names are associated with significantly higher death rates. An explanation for this unexpected finding is tested in six experiments. These experiments show that gender-congruent perceptions of intensity and strength are responsible for male-named hurricanes being perceived as riskier and more intense than female-named hurricanes. These findings have important implications for hurricane preparedness and public safety.

US hurricanes used to be given only female names, a practice that meteorologists of a different era considered appropriate due to such characteristics of hurricanes as unpredictability (17). This practice came to an end in the late 1970s with increasing societal awareness of sexism, and an alternating male-female naming system was adopted (17). Even though the gender of hurricanes is now preassigned and arbitrary, the question remains: do people judge hurricane risks in the context of gender-based expectations?

Research shows that women and men are socialized to have different social roles and self-schemas, in turn generating descriptive and prescriptive expectancies about women and men (16, 18). Men are often expected to be strong, competent, and aggressive, whereas women are often expected to be weak, warm, and passive (19–21). Men are more likely than women to commit violent behaviors (22), and thus males are perceived to be more strongly associated than females with negative potencies such as violence and destruction (23, 24). We extend these findings to hypothesize that the anticipated severity of a hurricane with a masculine name (Victor) will be greater than that of a hurricane with a feminine name (Victoria). This expectation, in turn, will affect the protective actions that people take. As a result, a hurricane with a feminine vs. masculine name will lead to less protective action and more fatalities.

Archival Study

To test this hypothesis, we used archival data on actual fatalities caused by hurricanes in the United States (1950–2012). Ninety-four Atlantic hurricanes made landfall in the United States during this period (25). Nine independent coders who were blind to the hypothesis rated the masculinity vs. femininity of historical hurricane names on two items (1 = very masculine, 11 = very feminine, and 1 = very man-like, 11 = very woman-like), which were averaged to compute a masculinity-femininity index (MFI). A series of negative binomial regression analyses (26, 27) were performed to investigate effects of perceived masculinity-femininity of hurricane names (MFI), minimum pressure, normalized damage (NDAM) (28), and the interactions among them on the number of deaths caused by the hurricanes (see *Materials and Methods* for complete descriptions of models tested, Table S1 for descriptive statistics, and Table S2 for a statistical summary of models tested. See the full Dataset S1 available online.)

The analyses showed that the change in hurricane fatalities as a function of MFI was marginal for hurricanes lower in normalized damage, indicating no effect of masculinity-femininity of name for less severe storms. For hurricanes higher in normalized damage, however, this change was substantial, such that hurricanes with feminine names were much deadlier than those

Significance

Meteorologists and geoscientists have called for greater consideration of social science factors that predict responses to natural hazards. We answer this call by highlighting the influence of an unexplored social factor, gender-based expectations, on the human toll of hurricanes that are assigned gendered names. Feminine-named hurricanes (vs. masculine-named hurricanes) cause significantly more deaths, apparently because they lead to lower perceived risk and consequently less preparedness. Using names such as Eloise or Charlie for referencing hurricanes has been thought by meteorologists to enhance the clarity and recall of storm information. We show that this practice also taps into well-developed and widely held gender stereotypes, with potentially deadly consequences. Implications are discussed for understanding and shaping human responses to natural hazard warnings.

Author contributions: K.J. and S.S. designed research; K.J. performed research; K.J. and J.M.H. analyzed data; K.J., S.S., and M.V. wrote the paper.

The authors declare no conflict of interest.

*This Direct Submission article had a prearranged editor.

[1]To whom correspondence may be addressed. E-mail: kjung4@illinois.edu or shavitt@illinois.edu.

This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1402786111/-/DCSupplemental.

Jung et al. (2014)

2.3 A multiverse himmicane analysis?

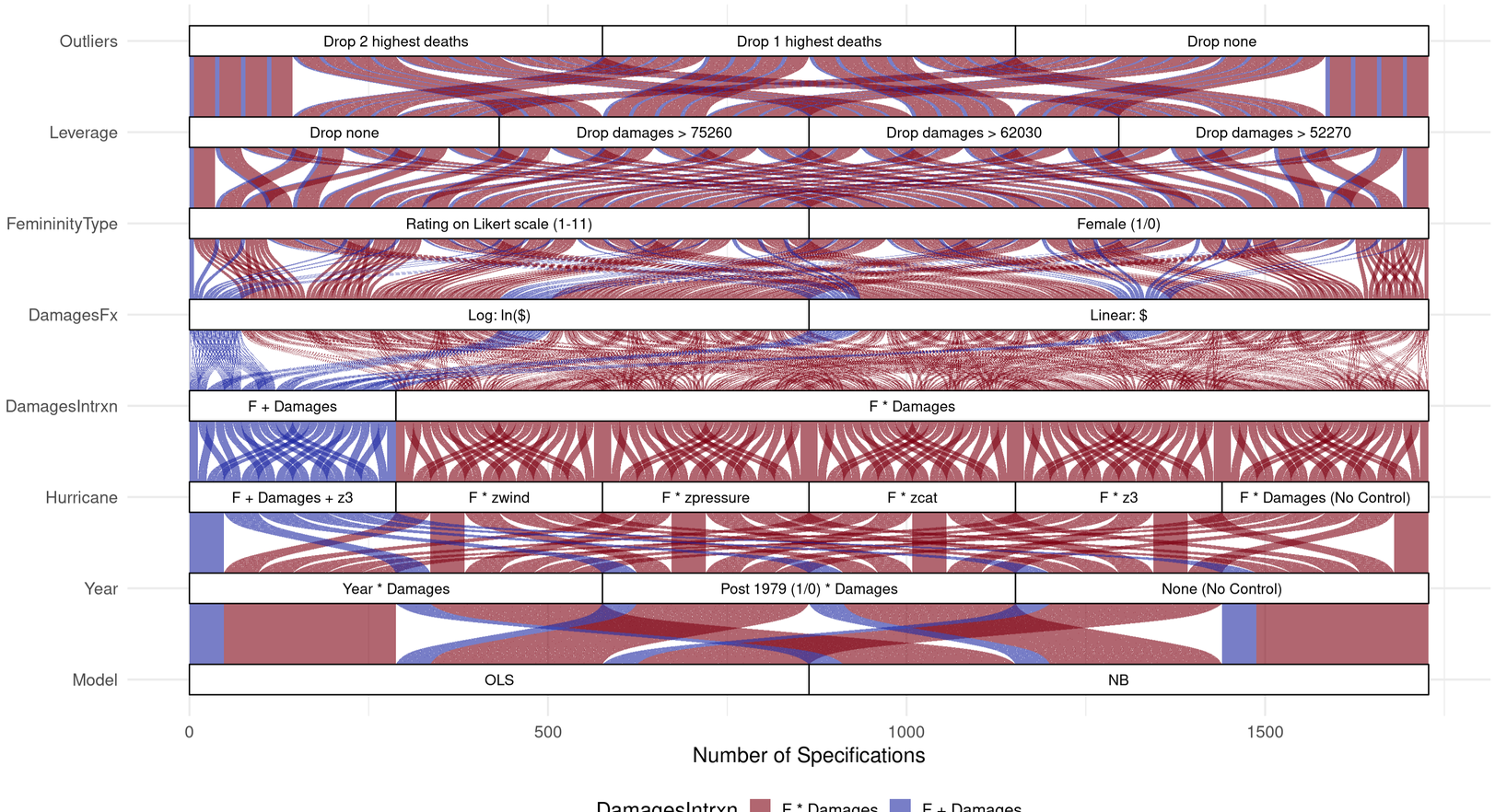


Figure 1: Coloured by different paths in the `DamagesIntrxn` branch (Simonsohn et al., 2020).

3 What are decision trees?

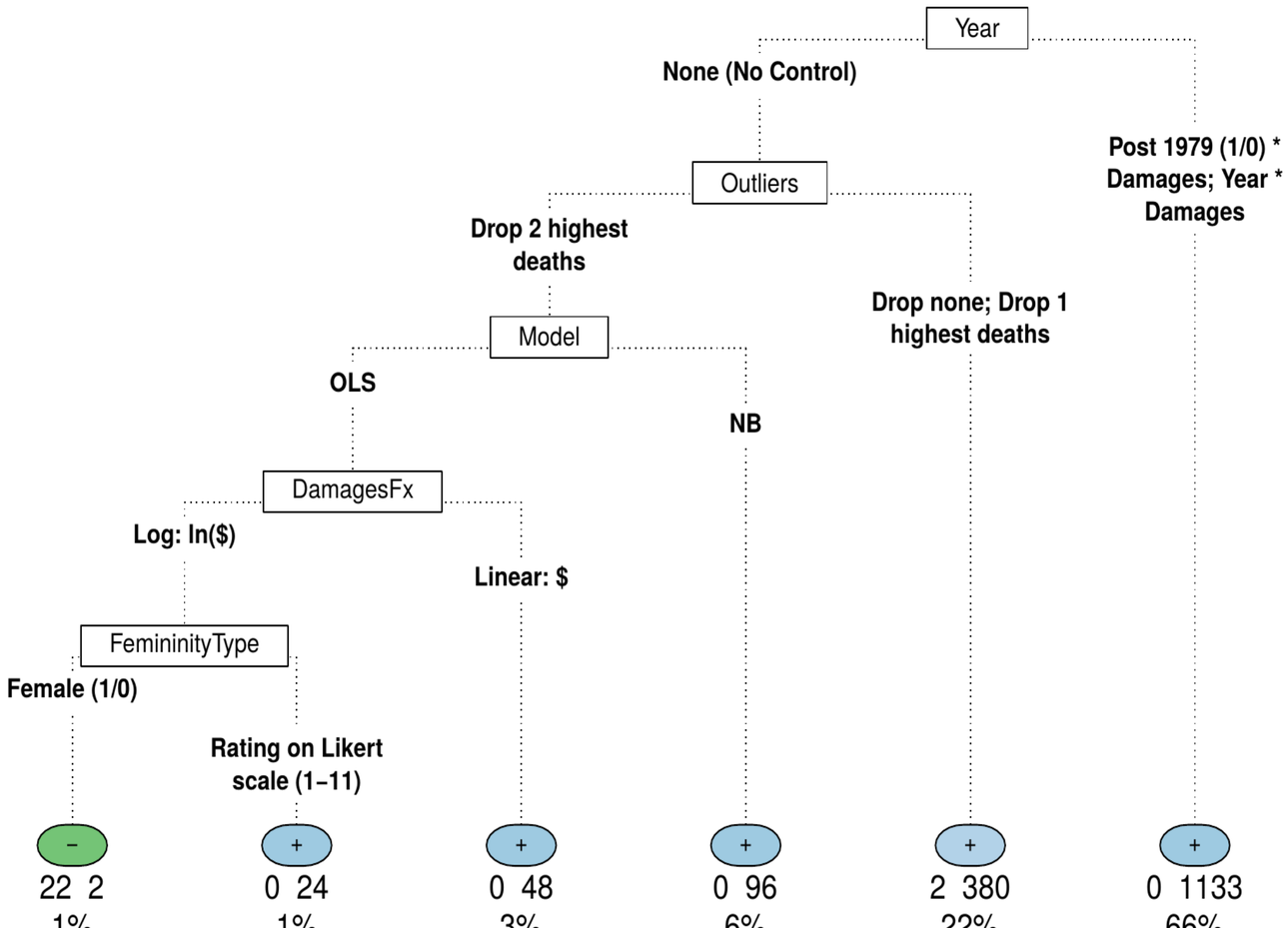
3.1 How can decision trees be applied to categorical data?

```
dt_sign <- rpart::rpart(
    difference_sign ~ Model +
    Year +
    Hurricane +
    DamagesIntrxn +
    DamagesFx +
    FemininityType +
    Leverage +
    Outliers,
    data = dt_dat
  )
```

99.77% of the sign of the differences was correctly predicted

Therefore, using this model, I would only be wrong about the sign of a multiverse analysis 0.23% of the time

- The tree tells us which combination of decisions *caused* a negative result and we can perform post-multiverse analysis investigations into *why*



Correctly Predicts Sign of Result 99.77% of the Time

4 What do decision trees tell us about multiverse analyses?

- Causal Inference
- How does this compare against Simonsohn et al. (2020)?
- Conclusion?
- Future?

4.1 Causal Inference of Multiverse Analysis?

- Is my tree big because the multiverse analysis is causally messy or because decisions moderate each other?
  - DAGs show the way
- DAGs also show when basic decision trees are inappropriate

4.2 Example of Complete Independence and DAG

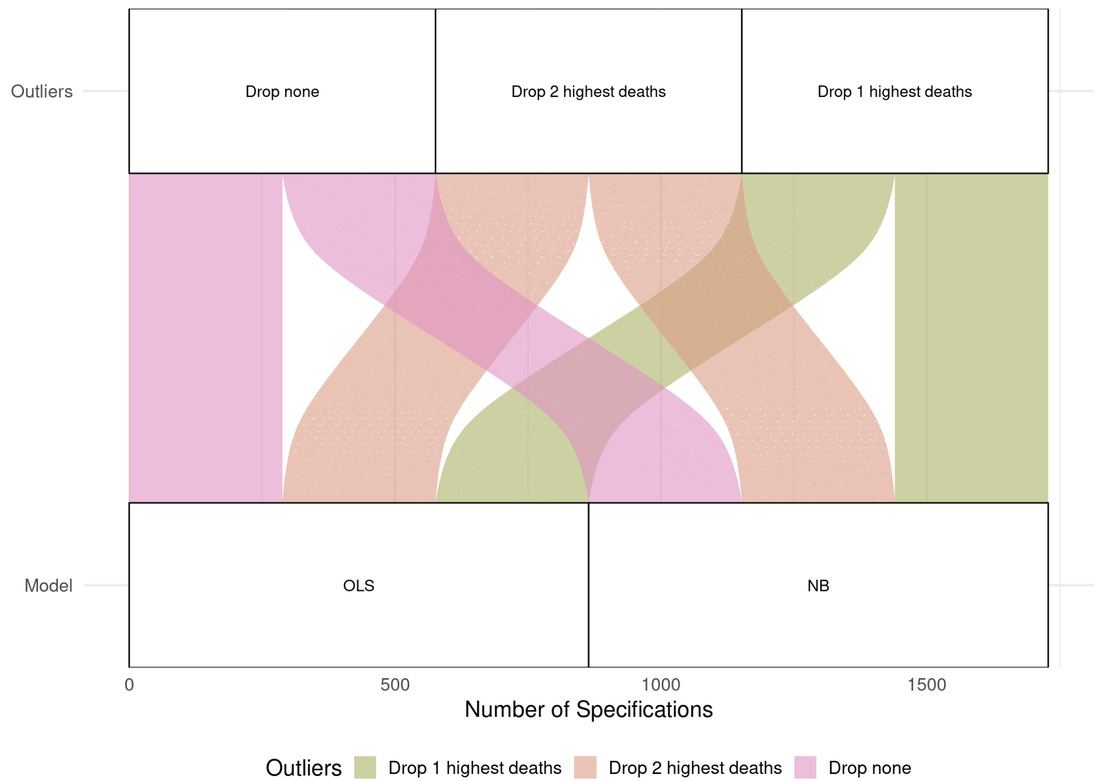


Figure 2

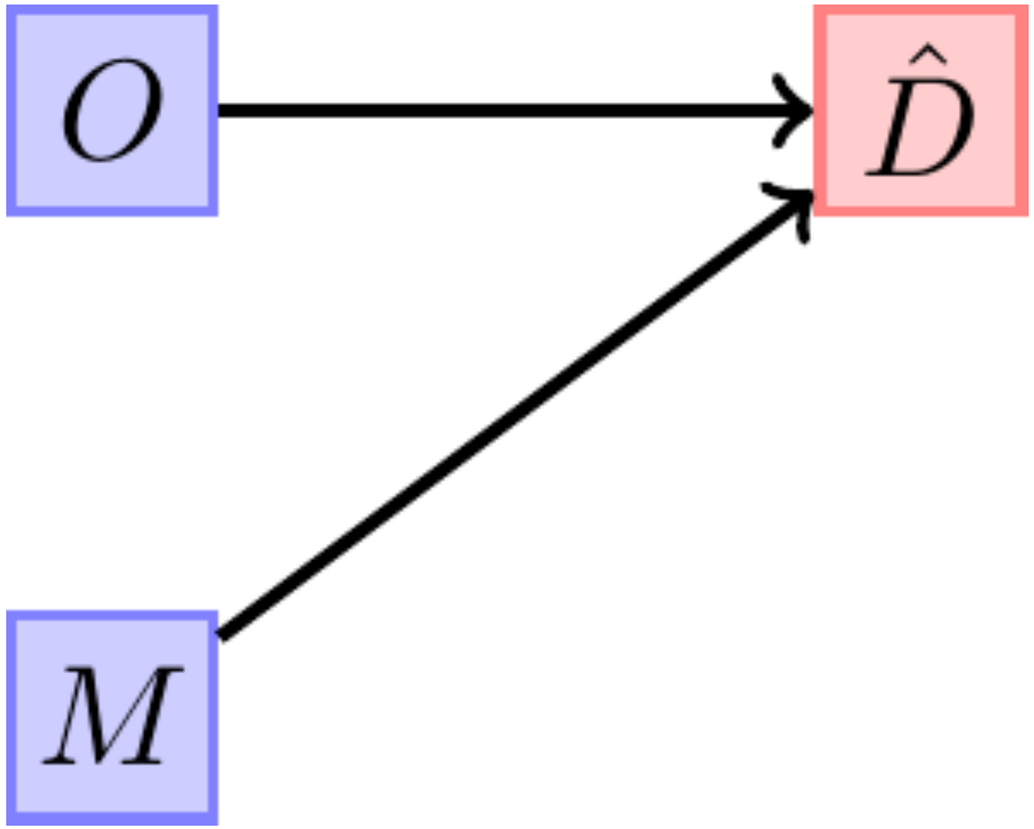


Figure 3: DAG of Complete Independence.

4.3 Example of Complete Mediation and DAG

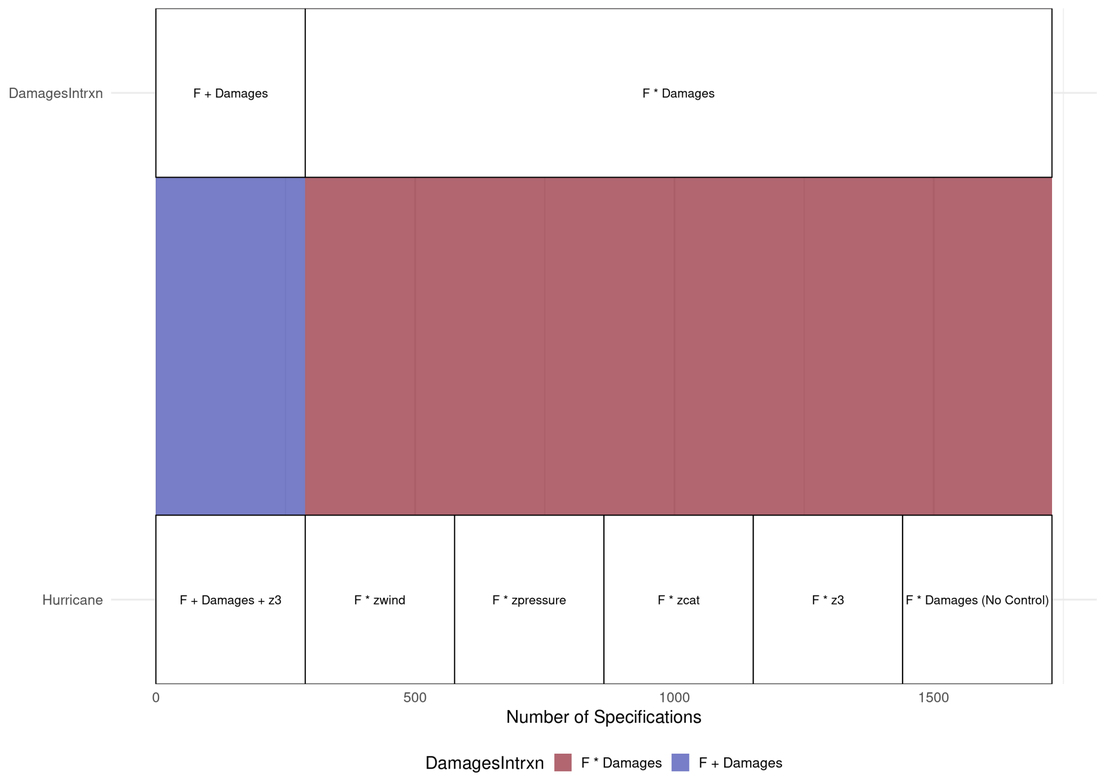


Figure 4

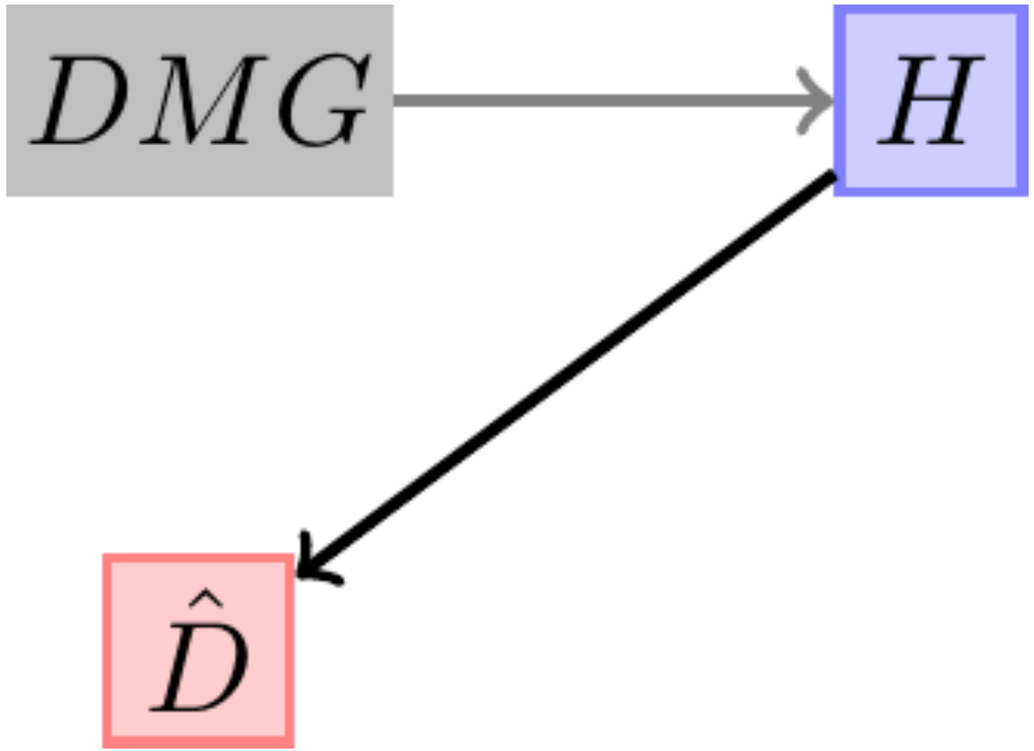


Figure 5: DAG of Complete Mediation

4.4 What does the DAG of the Himmicane multiverse analysis look like?

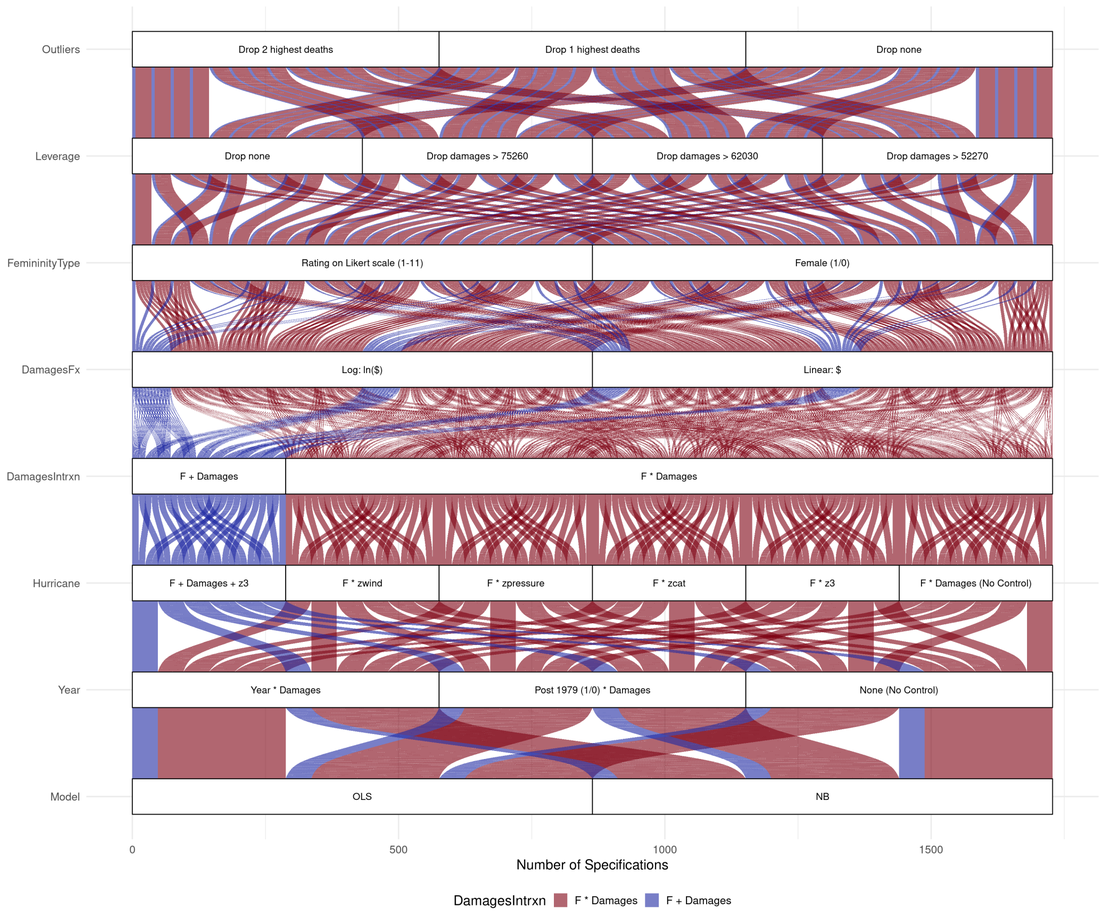


Figure 6

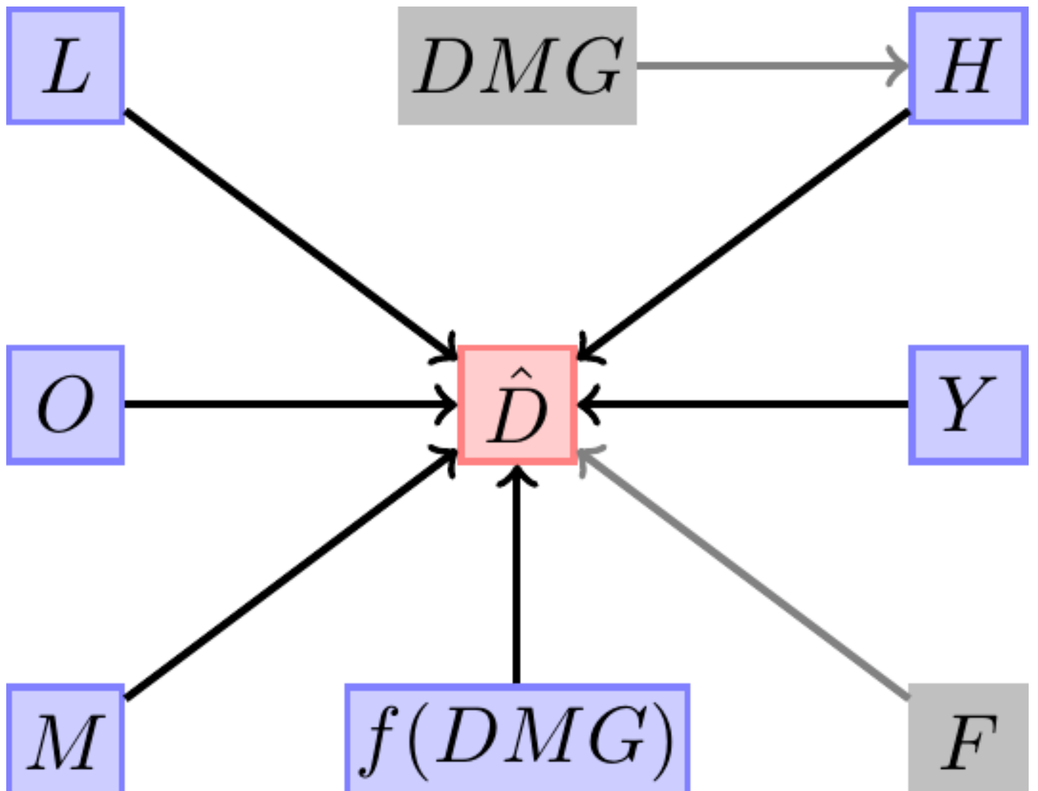
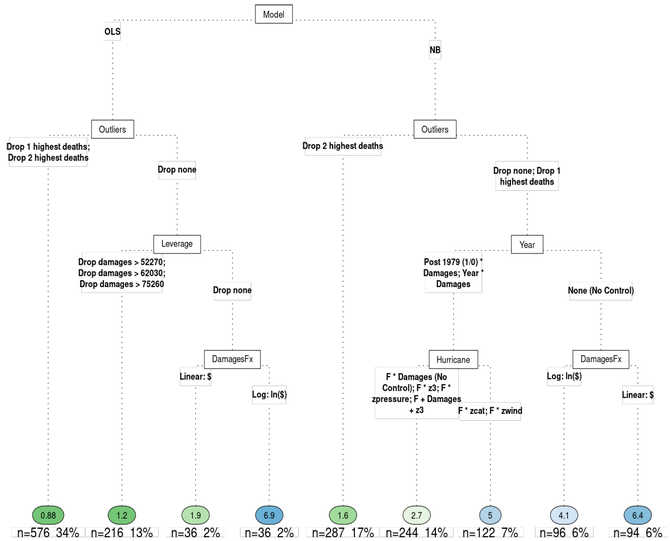


Figure 7: DAG of our Multiverse Hurricane Analysis. The node O stands for the 'Dropping Outliers' branch; L for 'Dropping Leverage Points'; DMG for 'Controlling for Damages'; H for 'Controlling for Hurricane Severity'; Y for 'Controlling for Year'; F for 'Femininity of Name'; $f(DMG)$ for 'Functional Form of Damages'; and M for 'Statistical Model'. Gray nodes and edges represent effects not captured by the regression decision tree.

5 How does this compare against Simonsohn et al. (2020)?

5.1 Difference Analysis

> “the treatment of outliers is consequential but the definition of hurricane severity less so; […] effects become larger as outliers are retained; and […] the negative binomial gives systematically larger results than log(deaths + 1)” (p.4-5)



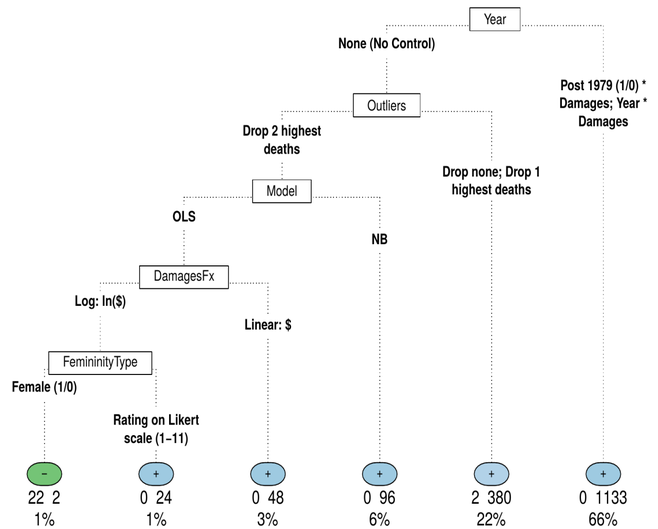
Decision Tree Difference Multiverse

In contrast,

- Although the treatment of outliers is predictive, controlling for hurricanes seems to only predict variation in the outcomes in the `NB` regression path
- Using `OLS` has even larger marginal effect than `NB` when:
  - No outliers are dropped;
  - No leverage points are dropped;
  - Damage predictor is log-transformed
  - Year of hurricane is controlled for

5.2 Sign Analysis

> “only by logging both damages and fatalities and dropping the two outliers does a sign reversal arise” (p.4)



Decision Tree Sign Multiverse Showing Selectivity of Results

In contrast,

- sign reversal occurs almost always when:
  - logging damages;
  - logging fatalities (meaning using the `OLS` path);
  - dropping the two highest outliers in deaths;
  - using a categorical indicator for Femininity;
  - and not controlling for Year.

5.3 Conclusion

Our Position:

1. Multiverse analyses should always be conducted and transparently reported to combat cherry-picking
2. Our framework helps to interpret multiverse reports

5.3.1 Beyond cherry-picking and honesty, why should I conduct a multiverse analysis?

- *Post-multiverse analysis judgment* ≠ *pre-multiverse analysis judgment* of equivalence of statistical approaches
- *Post-multiverse analysis judgment* ≠ *pre-multiverse analysis judgment* of equivalence of estimands
  - There are both obvious and non-obvious ways for the estimand to change with your analytical approach
  - Whether or not the estimand has changed can become evident after a multiverse analysis
  - What is and is not the estimand is heavily driven by substantive theory **and not statistics** per-se.

5.4 What are future extensions of this work?

- Multivariate (means) decision trees
- SEM trees - Come see my second talk for a related topic!
- Network trees

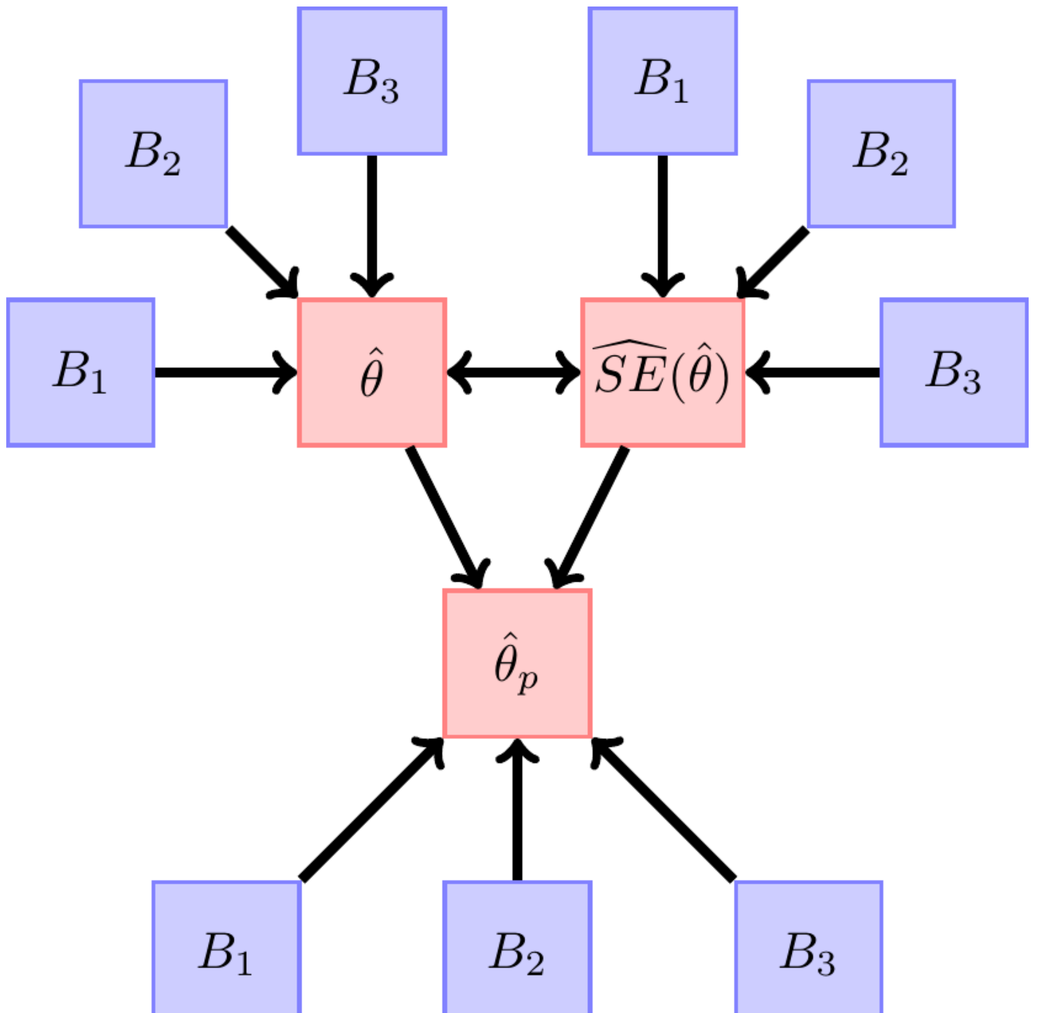


Figure 8

emstruonger@gmail.com

6 Questions?

Thank you for your time!

(Many extra features and perpsectives in our manuscript! e.g., What of Del Giudice and Gangestad (2021)?)

emstruonger@gmail.com

7 Bonus Content

7.1 Partial mediation example and DAG

WORK IN PROGRESS! POTENTIALLY INACCURATE

- It probably did not happen in our multiverse analysis
- Might happen when a subset of the paths in a downstream branch occur conditionally on a subset of the paths of an upstream branch

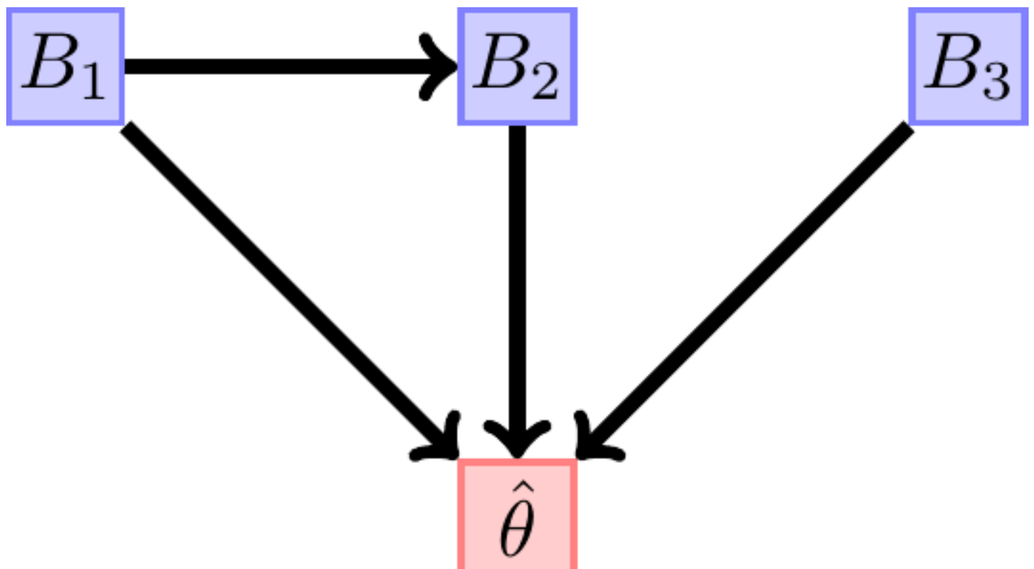
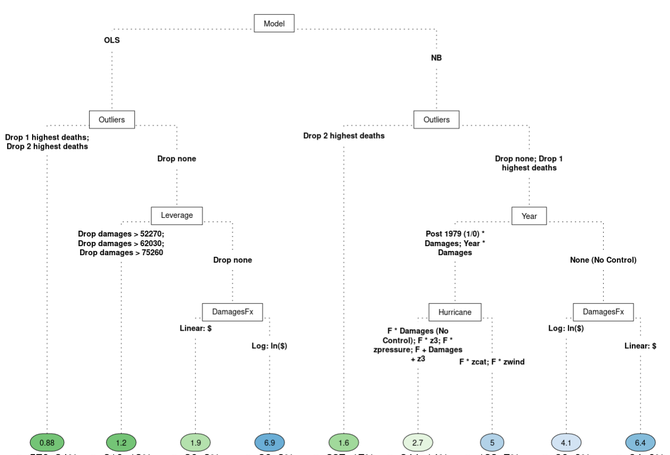


Figure 9: DAG of Partial Mediation

7.2 How do you run a decision tree analysis? How does it compare against regression?

Decision Tree Analysis ($R^2 = 0.707$)

```
library(rpart); library(rpart.plot)
dt <- rpart::rpart(difference ~ Model + Year +
                     Hurricane + DamagesIntrxn +
                     DamagesFx + FemininityType +
                     Leverage + Outliers,
                   data = dt_dat,
                   control = rpart.control(maxdepth = 4))
# maxdepth = 4 -- pdf size!
```



Decision Tree Analysis of the Difference Between Feminine and Masculine-Named Hurricanes

Compared with OLS ($R^2 = 0.484$)

```
ols <- lm(
  difference ~ Model +
    Year +
    Hurricane +
    DamagesIntrxn +
    DamagesFx +
    FemininityType +
    Leverage +
    Outliers,
  data = dt_dat
)
```

$$\text{difference} = \beta_0 + \beta_1(\text{Model}_{\text{OLS}}) + \beta_2(\text{Year}_{\text{Post 1979 (1/0)} * \text{Damages}}) + \beta_3(\text{Year} * \text{Damages}) + \beta_4(\text{Hurricane}_{\text{F} * \text{z3}}) + \beta_5(\text{Hurricane}_{\text{F} * \text{zcat}}) + \beta_6(\text{Hurricane}_{\text{F} * \text{zpressure}}) + \beta_7(\text{Hurricane}_{\text{F} * \text{zwind}}) + \beta_8(\text{Hurricane}_{\text{F + Damages + z3}}) + \beta_9(\text{DamagesIntrxn}_{\text{F} + \beta_{10}(\text{DamagesFx}_{\text{Log}} \times \text{DamagesFx}_{\text{ln(\$)}}) + \beta_{11}(\text{Femin} \beta_{12}(\text{Leverage}_{.\text{L}}) + \beta_{13}(\text{Leverage}_{.\text{Q}}) + \beta_{14}(\text{Leverage}_{.\text{C}}) + \beta_{15}(\text{Outliers}_{.\text{L}}) +$$

8 References

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